UPWORK WORK SAMPLE | LABELBOX JOB: LBLX116 - PROGRAMMERS FOR AI TRAINING

AI HALLUCINATION HUNTER

"Evidence-Grounded AI Claim Evaluation"

Asynchronous REST API * Atomic Claim Decomposition * Second-Pass Substring Entailment Verification

GITHUB REPOSITORY [CLICK TO OPEN]
github.com/sapnajha757/hallucination-hunter

Python 3.11+ | FastAPI | Pydantic v2 | 43 Passing Tests

PROJECT OVERVIEW & OBJECTIVE

AI Hallucination Hunter is an asynchronous Python and FastAPI evaluation service that assesses factual assertions in AI-generated answers against caller-provided reference evidence, emitting strongly-typed, structured JSON responses via Pydantic v2.

The service decomposes candidate answers into atomic, independently verifiable claims, checks each against an ingested evidence store, audits referential citation integrity, executes a focused second-pass semantic re-verification with exact literal quote validation, and computes deterministic hallucination risk scores.

CORE TECHNOLOGY STACK

Language & Runtime:	Python 3.11+ / Strict Async Execution
REST API Layer:	FastAPI >=0.110, Uvicorn ASGI Server
Data Contracts:	Pydantic v2 & Pydantic-Settings
LLM Orchestration:	OpenAI SDK (Async, Groq/Ollama ready)
Automated Tests:	Pytest & Pytest-Asyncio (100% Offline Mocks)

KEY ENGINEERING FEATURES & DEFENSE-IN-DEPTH ARCHITECTURE

1. Claim-Level Verdicts & Composite Risk Scoring

Decomposes candidate answers into atomic claims evaluated as SUPPORTED, CONTRADICTED, or UNVERIFIED. Emits bounded risk scores: Risk = (Contradictions x 1.0 + Unverified x 0.5) / Total Claims, mapped deterministically to SAFE, CAUTION, and HIGH_RISK qualitative statuses.

2. Second-Pass Semantic Reverification & Quotes

Targets SUPPORTED and CONTRADICTED claims in isolated XML boundaries. Submits discrete (claim, cited_passage) pairs and enforces that returned evidence quotes exist as exact literal substrings (quote in passage.text), eliminating semantic false positives and hallucinated citation IDs.

3. Fail-Safe Exception Boundaries & Degradation

Guarantees operational resilience across provider volatility. Upstream timeouts (HTTP 504), malformed model outputs (HTTP 502), ungrounded quotes, or second-pass disagreements safely downgrade claims to UNVERIFIED with confidence 0.0, recalculating risk metrics without pipeline crashes.

4. Security Hardening & Threat Mitigation

Defense-in-depth against prompt injection: user inputs are isolated in XML containers; structural tags (</candidate_answer>, </evidence>) are escaped case-insensitively (</...>). Enforces strict regex on evidence IDs (^[a-zA-Z0-9_\-\.:]+$) and provides sanitized 500 error shields.

DEMONSTRATION SCENARIOS: REALISTIC PIPELINE OUTCOMES

VERDICT: SUPPORTED

SAFE * Risk: 0.0 * Conf: 0.98

Candidate Claim:
"The Eiffel Tower was completed in 1889 for the World's Fair."

Reference Evidence:
Encyclopaedia Britannica: 'Constructed from 1887 to 1889 as the centerpiece of the 1889 World's Fair.'

Audit: Evidence agrees; second-pass entailment confirmed and quote validated as exact literal substring. Safe risk score (0.0).

VERDICT: CONTRADICTED

HIGH_RISK * Risk: 1.0 * Conf: 0.99

Candidate Claim:
"The Eiffel Tower was completed in 1999 for the World's Fair."

Reference Evidence:
Encyclopaedia Britannica: 'Constructed from 1887 to 1889 as the centerpiece of the 1889 World's Fair.'

Audit: Direct factual conflict: candidate says 1999, while evidence states 1889. Marked CONTRADICTED; triggers immediate HIGH_RISK status.

VERDICT: UNVERIFIED

CAUTION * Risk: 0.25 * Conf: 0.00

Candidate Claim:
"The Phoenix-7 flagship delivers up to 48 hours continuous battery life."

Reference Evidence:
Hardware Datasheet: 'The Phoenix-7 features an octa-core neural processor built on 3nm architecture.'

Audit: Unrelated evidence: battery-life claim is paired with unrelated evidence about an octa-core processor. Safely downgraded to UNVERIFIED (0.0).

RELEVANCE TO LABELBOX (LBLX116 ROLE)

* **Analytical Thinking & Fact-Checking:**
  Deconstructs nuanced AI reasoning into discrete atomic units. Focuses on rigorous evidence-based factuality, identifying subtle hallucinations, unsupported generalizations, and factual contradictions.
* **Evidence-Grounded AI Evaluation:**
  Evaluates model claims strictly relative to provided reference corpora, refusing to import outside knowledge or assume unverified plausibility when ground-truth reference data is silent.
* **Adversarial Testing & Edge Cases:**
  Anticipates real-world failure modes including subtle citation fabrication, prompt injection attempts, malformed schema outputs, and misleading quotes taken out of context.
* **Reliable Output Engineering:**
  Delivers structured Pydantic data models, clean asynchronous architectures, and deterministic scoring algorithms essential for high-fidelity AI training

AUTOMATED TESTING & REPORTED RESULTS

REPORTED RESULT: 43 PASSING TESTS

The project has previously reported 43 passing tests. (Reported result, not a newly verified test

Test Suite Coverage Breakdown (100% Offline Mocks):

* **Core Endpoints & OpenAPI:**
  Validates /hunt, /health service metadata, and Swagger schema specs.
* **Scoring Logic & Edge Cases:**
  Examines formula edge cases, conflicting citations, and zero-claim handling.
* **Payload & DoS Boundaries:**
  Enforces strict limits on characters (50k total) and evidence counts (20 max).
* **Prompt Injection Security:**
  Validates case-insensitive tag escaping (</evidence> to </evidence>).
* **Citation Fabrications:**
  Strips fabricated evidence IDs, automatically degrading claims to UNVERIFIED.
* **Second-Pass Fail-Safe**

PROJECT LIMITATIONS & EPISTEMIC DISCLAIMER

* Grounding vs. Objective Truth: AI Hallucination Hunter assists with evidence-grounded evaluation against caller-provided texts; it cannot guarantee absolute universal truth or replace expert human review. If user-supplied evidence contains errors or omissions, evaluations strictly reflect those source limitations.

* Quote Matching Caveat: Exact literal substring matching confirms that cited text fragments physically appear in reference passages, preventing citation fabrication. However, literal quote matching alone does not prove semantic entailment; quotes can be selectively extracted, ironic, or contextually inverted.

* Heuristic Nature: Confidence scores represent heuristic measures of semantic alignment, not calibrated probabilities. Automated evaluation serves as an efficient verification assist layer alongside human-in-the-loop oversight.